Fast Axiomatic Attribution for Neural Networks

Robin Hesse[1] Simone Schaub-Meyer[1] Stefan Roth[1,2]

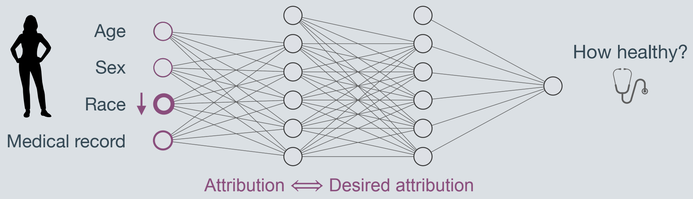
[1] TECHNISCHE UNIVERSITÄT DARMSTADT [2] hessian.AI

NEURAL INFORMATION PROCESSING SYSTEMS

Page & Code

Motivation

By enforcing priors on the feature attribution of a NN we can actively control the network's behavior [3], e.g., to correct for biases present in the training data [2].



Attribution should be **axiomatic** *and* **efficiently computable** to be practical [1]:

Integrated Gradients [5]	$\mathcal{X}$-Gradient	Input × Gradient [4]
Axiomatic ✓	Axiomatic ✓	Axiomatic ✗
Efficient ✗	Efficient ✓	Efficient ✓
	Our method	

Contributions

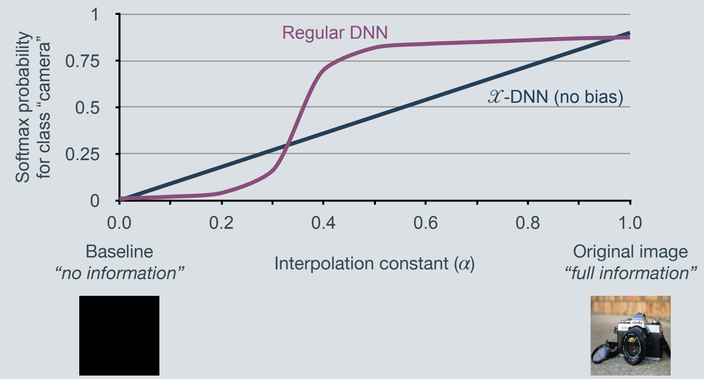
- We show that for nonnegatively homogeneous DNNs (here termed $\mathcal{X}$-DNNs) we can compute closed-form axiomatic feature attributions [5] with only a single forward/backward pass (20-300 times more efficient than for regular DNNs)
- We establish a new theoretical connection between the attribution methods Input × Gradient [4] and Integrated Gradients [5] for nonnegatively homogeneous DNNs of different degrees
- We show how $\mathcal{X}$-DNNs can be instantiated from a wide range of regular DNNs by simply removing the bias term, which has a surprisingly minor impact on the predictive accuracy
- We beat state-of-the-art generic attribution methods for training regular networks with an attribution prior

References & disclosure of funding

[1] Erion et al. (2021). "Improving performance of deep learning models with axiomatic attribution priors and expected gradients." In: Nature Machine Intelligence
[2] Obermeyer et al. (2019). "Dissecting racial bias in an algorithm used to manage the health of populations." In: Science
[3] Ross et al. (2017). "Right for the Right Reasons: Training Differentiable Models by Constraining their Explanations." In: IJCAI
[4] Shrikumar et al. (2016). "Not just a black box: Learning important features through propagating activation differences." In: arXiv:1605.01713 [cs.LG]
[5] Sundararajan et al. (2017). "Axiomatic attribution for deep networks." In: ICML

This project has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (grant agreement No. 866008). The project has also been supported in part by the State of Hesse through the cluster projects "The Third Wave of Artificial Intelligence (3AI)" and "The Adaptive Mind (TAM)".

Efficiently axiomatically attributable DNNs



Integrated Gradients (IG) [5] computes the integral of the gradient of the softmax probability of a DNN by interpolating between a neutral baseline and the input:

$$IG(F, input, baseline) = (input - baseline) \int_0^1 \bigtriangledown F(\alpha \times input + (1-\alpha) \times baseline)\ d\alpha$$

Issue: For a regular DNN, each point of the softmax probability function (purple) requires a separate forward pass, and thus, IG is computed via an expensive numerical approximation.

Solution: For our $\mathcal{X}$-DNN, this function is linear (blue) and a closed-form solution of IG is given by our novel axiomatic $\mathcal{X}$-Gradient attribution method:

$$\mathcal{X}G(F, input) = IG(F, input, \mathbf{0}) = input \times \bigtriangledown F(input)$$

Consequence: IG can be computed with $\mathcal{X}$-Gradient for $\mathcal{X}$-DNNs with only one forward/backward pass compared to 20-300 passes needed for regular DNNs.

→ **Two orders of magnitude more efficient ⟹ suited for training**

Obtaining $\mathcal{X}$-DNNs

Nonnegative homogeneity: $\forall \alpha \in \mathbb{R}_{\geq 0} : \alpha F(x) = F(\alpha x)$.

A DNN is nonnegatively homogeneous ($\mathcal{X}$-DNN) if all of its components are nonnegatively homogeneous.

Component	Nonnegatively homogeneous
Pooling	✓
ReLU	✓
Linear	✗
Convolution	✗

Linear
$$\alpha(Wx + b) = W\alpha x + \boxed{\alpha b} \overset{!}{\neq} W(\alpha x) + \boxed{b}$$

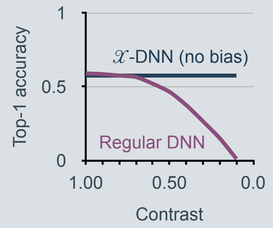
→ **Remove the bias term from ReLU DNNs to obtain $\mathcal{X}$-DNNs**

Experimental results

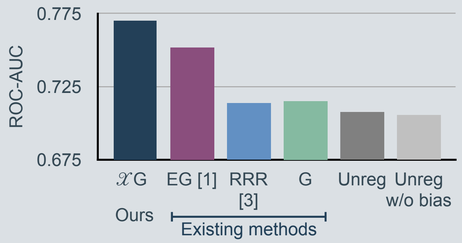
- Removing the bias term impairs accuracy only marginally:

Model	Top-5 accuracy (ImageNet) AlexNet	VGG16	ResNet-50
Regular DNN	**79.21**	**90.44**	**92.56**
$\mathcal{X}$-DNN (no bias)	78.54	90.25	91.12

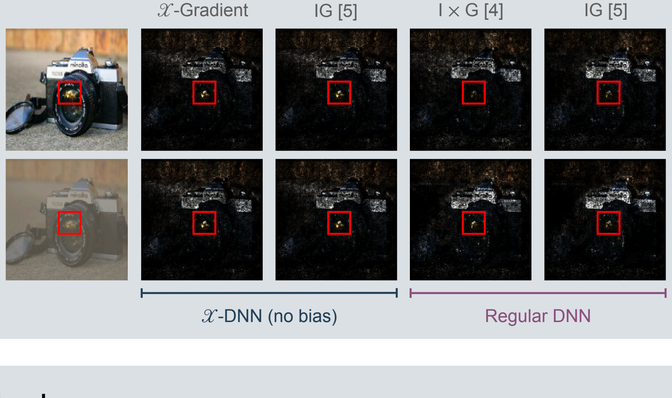
- Nonnegative homogeneity of $\mathcal{X}$-DNNs leads to contrast invariance



- $\mathcal{X}$-Gradient beats state-of-the-art attribution methods for training with an attribution prior



- For $\mathcal{X}$-DNNs:
  - $\mathcal{X}$-Gradient attribution equals Integrated Gradients
  - Reducing contrast doesn't change the attribution (up to scaling factor)



Take home

- For nonnegatively homogeneous DNNs ($\mathcal{X}$-DNNs), Input × Gradient [4] equals Integrated Gradients [5]
- Remove the bias term from ReLU DNNs to obtain $\mathcal{X}$-DNNs
- Use $\mathcal{X}$-DNNs when you need efficiently computable axiomatic attributions, e.g., for training with attribution priors
- Use $\mathcal{X}$-DNNs for applications that require equivariance to contrast changes

https://visinf.github.io/fast-axiomatic-attribution